

NLP - 2026, Quiz - 1

1. Identify the predicate in the following sentences, if any. **[3 marks]**
- The man who hunts **ducks** out on weekends.
 - The old **man** the boat.
 - The complex **houses** married and single soldiers and their families.

2. Highlight the differences between Bengio's Neural Language Model and Mikolov's Continuous Bag-of-words architectures. **[3 marks]**

Note: We are not looking for diagrams.

Answer:

- Objectives: Predict the next word vs predict the center word
- Context: Only previous tokens as context vs both previous and future tokens as context.
- Hidden layer: Non-linear (tanh) vs linear projection

Note: Any two would be fine. 1.5 marks each.

3. Mr X is designing an NER system and asks his assistant to process the dataset with an appropriate encoding scheme. Additionally, he mentioned that two or more consecutive entities will always be delimited by some punctuation or conjunctions. Suggest the most appropriate scheme the assistant should employ and why? **[2 marks]**

The most appropriate scheme: Either one of them.

Coarse-grained: IO encoding (two labels)

OR

Fine-grained: Person, Location, Organization, and OTHER

Note: The other class is important. No marks without it.

Reason: Since it is guaranteed that a delimiter (punctuation) or conjunctions (and/or) will come in between two or more entities, delimiter/conjunctions will take the OTHER class.

Virat	Kohli	and	Sachin	are	Indian	cricketers	.	
I	I	O	I		O	I	O	
Per	Per	O	Per		O	Loc	O	

Note: BIO scheme will not fetch any mark. Computationally, binary clarifications (i.e., IO) will be easier than ternary classification (i.e., BIO).

4. Justify the use of negative sampling in Mikolov's word2vec model. **[2 marks]**
- A word usually has a small set of relevant / contextual words; however, it has a huge number of words (remaining words) as irrelevant / non-contextual words. To compute the likelihood of contextual word w_t given the current word w_c , the original word2vec computes the similarity (dot product) with the contextual word as numerator and normalizes it over all other words (including irrelevant words) in the vocabulary as denominator – a costly operation. To reduce the complexity of the system, negative sampling uses a small-and-fixed set of negative / non-contextual words w.r.t. the current word and maximize the likelihood for contextual words and minimize for the non-contextual words.
5. State the differences between the byte-pair encoding and the word-piece tokenizer. **[2 marks]**
- BPE merges the most frequent tokens, whereas, word-piece merges tokens that maximized the likelihood of the data.
6. Describe FastText Word Representations and give an example of fasttext input. **[3 marks]**
- Fasttext word representations are computed using word2vec models at the subword (or character n-gram) level. For each token, it computes character n-grams for (usually, $n = 3$ to 6) and merges all of them including the original token as an input to the fasttext model.

Example: For the token "where", input will be:

[whe, her, ere, wher, here, where]